Counts, laws and denominators

Fix report g and information y; use nonnegative weights summing to one.
Choose ν = s (supplied) or P (evaluation).

$$M = \binom{n}{2} > 0, \quad x_{ij} = \mathbf{1}\{g_i < g_j\}, \quad N(g) = \sum_{i \neq j} x_{ij}$$

$N(g)/M$ is raw disclosure; it is fixed by g.

Realized count	Average under ν at y	Rate and condition
Weak error (≤)		
$D_k(g, \Theta) = \sum_{i \neq j} x_{ij} \mathbf{1}\{\theta_{ik} \leq \theta_{jk}\}$	$E_k^\nu(g) = \mathbb{E}_\nu[D_k \mid y]$	$E_k^\nu(g)/N(g)$ $N(g) > 0$
Strict reversal (<)		
$\sum_{i \neq j} x_{ij} \mathbf{1}\{\theta_{ik} < \theta_{jk}\}$	$E_k^{b,\nu}(g)$ Expected strict-error count	$E_k^{b,\nu}(g)/N(g), \ N(g) > 0$ $E_k^{b,\nu}(g)/Q_k^\nu(g)$ $\quad Q_k^\nu(g) > 0$
Distinct opportunity (≠)		
$\sum_{i \neq j} x_{ij} \mathbf{1}\{\theta_{ik} \neq \theta_{jk}\}$	$Q_k^\nu(g)$ Expected distinct mass	$Q_w^\nu(g)/T_w^\nu$ $T_w^\nu > 0$

Weighting and total opportunity

$$Q_w^\nu(g) = \sum_k w_k Q_k^\nu(g), \quad T_w^\nu = \sum_{i<j} \sum_k w_k d_{ij,k}^\nu$$

$d_{ij,k}^\nu = \Pr_\nu(\theta_{ik} \neq \theta_{jk} \mid y)$

T: all original pairs; Q: disclosed directions.

Inspected tie_aware_v1 field names

API D ↔ $Q_w^s(g)$; API T ↔ T_w^s

Supplied-law aliases; book D_k remains a realized weak wrong-count.

Zero-denominator and varying-report qualifications follow in the text.